coloring

October 26, 2026


```
This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
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```


[1]:
```
import casadi as ca
```

Read all about coloring in the seminal paper "What color is your Jacobian?" http://www.cs.odu.edu/~pothen/Papers/sirev2005.pdf

[2]:
```
def color(A):
  print("="*80)
  print("Original:")
  print(repr(ca.DM(A,1)))
  print("Colored: ")
  print(repr(ca.DM(A.uni_coloring(),1)))
```

[3]:
```
A = ca.Sparsity.diag(5)
color(A)
```

```
================================================================================
Original:
DM(
```

```
[[1, 00, 00, 00, 00],
 [00, 1, 00, 00, 00],
 [00, 00, 1, 00, 00],
 [00, 00, 00, 1, 00],
 [00, 00, 00, 00, 1]])
Colored:
DM([1, 1, 1, 1, 1])
```

One direction needed to capture all

[4]:
```
color(ca.Sparsity.dense(5,10))
```

```
================================================================================
Original:
DM(
[[1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
 [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
 [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
 [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
 [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]])
Colored:
DM(
[[1, 00, 00, 00, 00, 00, 00, 00, 00, 00],
 [00, 1, 00, 00, 00, 00, 00, 00, 00, 00],
 [00, 00, 1, 00, 00, 00, 00, 00, 00, 00],
 [00, 00, 00, 1, 00, 00, 00, 00, 00, 00],
 [00, 00, 00, 00, 1, 00, 00, 00, 00, 00],
 [00, 00, 00, 00, 00, 1, 00, 00, 00, 00],
 [00, 00, 00, 00, 00, 00, 1, 00, 00, 00],
 [00, 00, 00, 00, 00, 00, 00, 1, 00, 00],
 [00, 00, 00, 00, 00, 00, 00, 00, 1, 00],
 [00, 00, 00, 00, 00, 00, 00, 00, 00, 1]])
```

We need 5 directions. The colored response reads: each row corresponds to a direction; each column correspond to a row of the original matrix.

[5]:
```
color(A+ca.Sparsity.triplet(5,5,[0],[4]))
```

```
================================================================================
Original:
DM(
[[1, 00, 00, 00, 1],
 [00, 1, 00, 00, 00],
 [00, 00, 1, 00, 00],
 [00, 00, 00, 1, 00],
 [00, 00, 00, 00, 1]])
Colored:
DM(
[[1, 00],
```

```
 [1, 00],
 [1, 00],
 [1, 00],
 [00, 1]])
```

First 4 rows can be taken together, the fifth row is taken separately

[6]:
```
color(A+ca.Sparsity.triplet(5,5,[4],[0]))
```

```
================================================================================
Original:
DM(
[[1, 00, 00, 00, 00],
 [00, 1, 00, 00, 00],
 [00, 00, 1, 00, 00],
 [00, 00, 00, 1, 00],
 [1, 00, 00, 00, 1]])
Colored:
DM(
[[1, 00],
 [1, 00],
 [1, 00],
 [1, 00],
 [00, 1]])
```

First 4 rows can be taken together, the fifth row is taken separately

[7]:
```
color(A+ca.Sparsity.triplet(5,5,[0]*5,list(range(5))))
```

```
================================================================================
Original:
DM(
[[1, 1, 1, 1, 1],
 [00, 1, 00, 00, 00],
 [00, 00, 1, 00, 00],
 [00, 00, 00, 1, 00],
 [00, 00, 00, 00, 1]])
Colored:
DM(
[[1, 00, 00, 00, 00],
 [00, 1, 00, 00, 00],
 [00, 00, 1, 00, 00],
 [00, 00, 00, 1, 00],
 [00, 00, 00, 00, 1]])
```

The first row is taken separately. The remainding rows are lumped together in one direction.

[8]:
```
color(A+ca.Sparsity.triplet(5,5,list(range(5)),[0]*5))
```

```
================================================================================
```

```
Original:
DM(
[[1, 00, 00, 00, 00],
 [1, 1, 00, 00, 00],
 [1, 00, 1, 00, 00],
 [1, 00, 00, 1, 00],
 [1, 00, 00, 00, 1]])
Colored:
DM(
[[1, 00],
 [00, 1],
 [00, 1],
 [00, 1],
 [00, 1]])
```

We need 5 directions.

Next, we look at star_coloring

[9]:
```
def color(A):
  print("="*80)
  print("Original:")
  print(repr(ca.DM(A,1)))
  print("Star colored: ")
  print(repr(ca.DM(A.star_coloring(1),1)))
```

[10]:
```
color(A)
```

```
================================================================================
Original:
DM(
[[1, 00, 00, 00, 00],
 [00, 1, 00, 00, 00],
 [00, 00, 1, 00, 00],
 [00, 00, 00, 1, 00],
 [00, 00, 00, 00, 1]])
Star colored:
DM([1, 1, 1, 1, 1])
```

One direction needed to capture all

[11]:
```
color(ca.Sparsity.dense(5,5))
```

```
================================================================================
Original:
DM(
[[1, 1, 1, 1, 1],
 [1, 1, 1, 1, 1],
 [1, 1, 1, 1, 1],
 [1, 1, 1, 1, 1],
```

```
 [1, 1, 1, 1, 1]])
Star colored:
DM(
[[1, 00, 00, 00, 00],
 [00, 1, 00, 00, 00],
 [00, 00, 1, 00, 00],
 [00, 00, 00, 1, 00],
 [00, 00, 00, 00, 1]])
```

We need 5 directions.

[12]:
```
color(A+ca.Sparsity.triplet(5,5,[0]*5,list(range(5)))+ca.Sparsity.
 ↪triplet(5,5,list(range(5)),[0]*5))
```

```
================================================================================
Original:
DM(
[[1, 1, 1, 1, 1],
 [1, 1, 00, 00, 00],
 [1, 00, 1, 00, 00],
 [1, 00, 00, 1, 00],
 [1, 00, 00, 00, 1]])
Star colored:
DM(
[[1, 00],
 [00, 1],
 [00, 1],
 [00, 1],
 [00, 1]])
```

The first row/col is taken separately. The remainding rows/cols are lumped together in one direction.

Let's take an example from the paper

[13]:
```
A = ca.
 ↪DM([[1,1,0,0,0,0],[1,1,1,0,1,1],[0,1,1,1,0,0],[0,0,1,1,0,1],[0,1,0,0,1,0],[0,1,0,1,0,1]])
A = ca.sparsify(A)
color(A.sparsity())
```

```
================================================================================
Original:
DM(
[[1, 1, 00, 00, 00, 00],
 [1, 1, 1, 00, 1, 1],
 [00, 1, 1, 1, 00, 00],
 [00, 00, 1, 1, 00, 1],
 [00, 1, 00, 00, 1, 00],
 [00, 1, 00, 1, 00, 1]])
Star colored:
```

```
DM(
[[00, 1, 00],
 [1, 00, 00],
 [00, 1, 00],
 [00, 00, 1],
 [00, 1, 00],
 [00, 1, 00]])
```